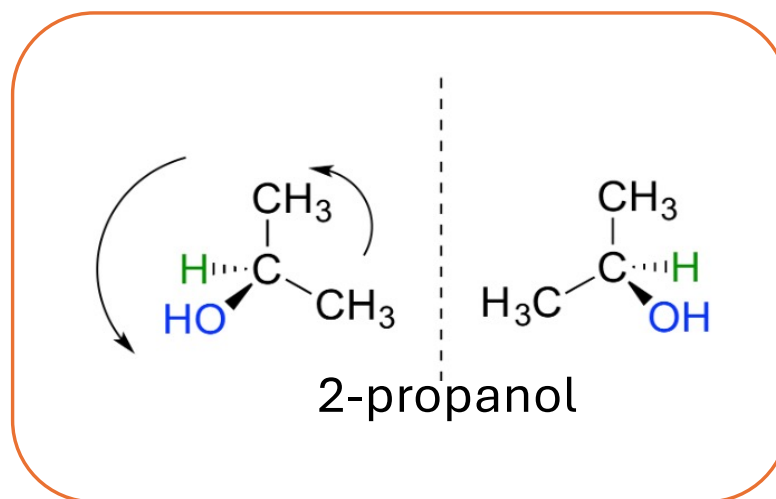
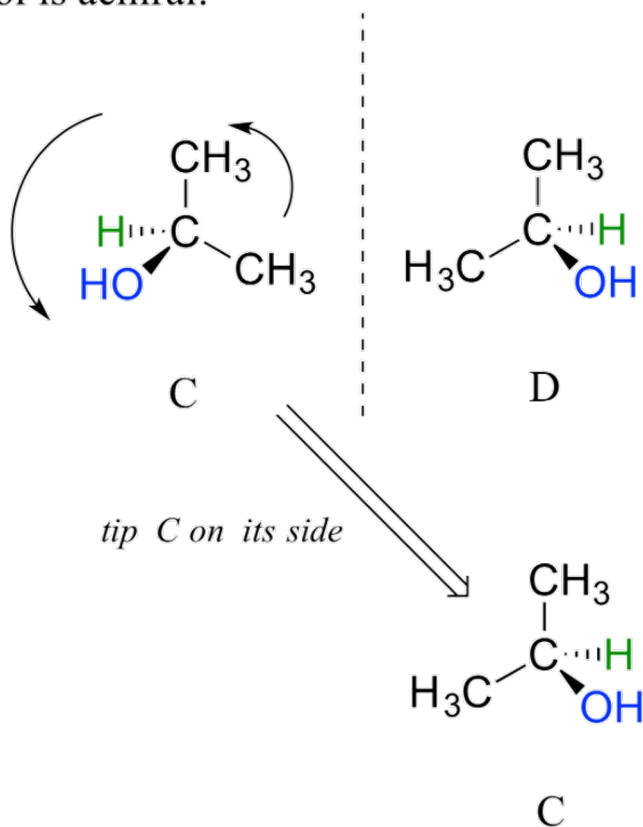


Test for chirality



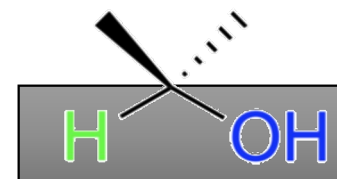
Can the mirror images be superimposed?
Search for symmetry elements
Chiral molecule? Enantiomeric?

2-propanol is achiral:



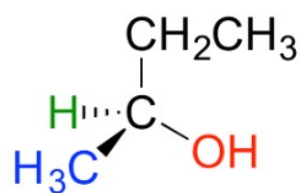
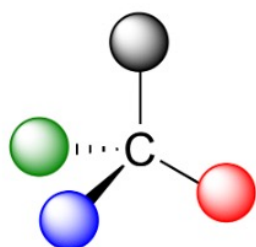
C and D can be superposed: they are the same molecule!

Plane of symmetry contains the atoms C, H and OH and passes through the center of the other two methyl groups



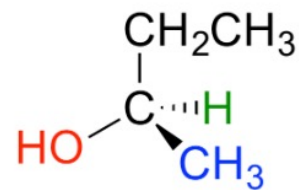
Justification is necessary

Test for chirality

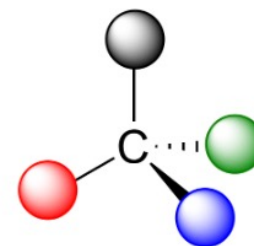


A

2-butanol



B



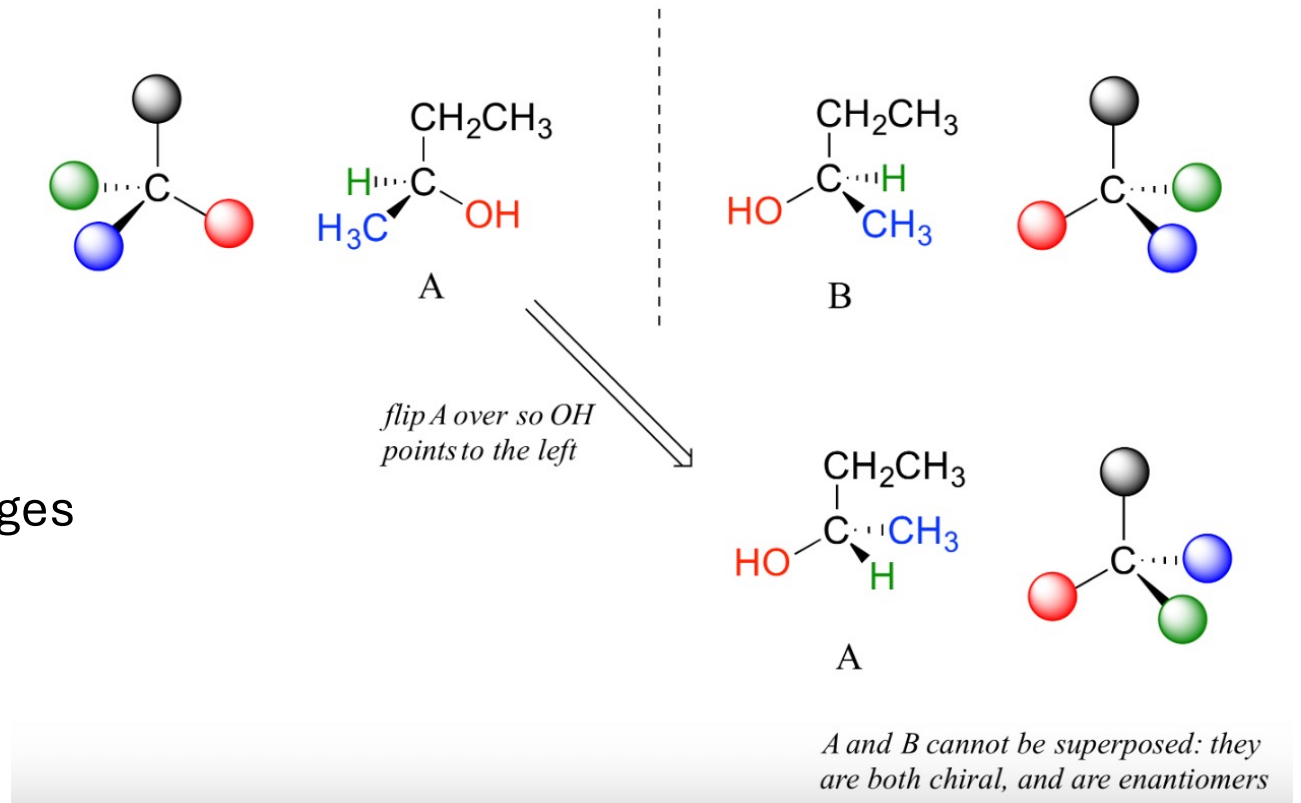
2-butanol is chiral

Presence of chiral carbon center (attached to four different groups)

No center of symmetry (i)

No plane of symmetry (σ)

Non-superimposable mirror images



Optical properties of enantiomers

Enantiomers interact differently with polarized light

Rotation of polarized light occurs – in both directions

Absolute value of rotation is fixed for enantiomers, and they are same

+ / - refers to rotation of light of enantiomers

The diagram shows the formula for specific rotation, $[\alpha]_{\lambda} = \frac{100\alpha}{cl}$, enclosed in a yellow rounded rectangle. Four blue arrows point from labels to parts of the formula: 'Specific rotation' points to $[\alpha]_{\lambda}$, 'Observed rotation' points to α , 'concentration' points to c , and 'Path length' points to l .

$$[\alpha]_{\lambda} = \frac{100\alpha}{cl}$$

Specific optical rotation depends on wavelength, temperature and solvent

Rotation is observed using a **polarimeter** (*vide ante*)

Homochirality

*Racemic
mixture*

*Enantiomeric
excess*

Depending on how it was obtained, a sample of a chiral compound can contain only one enantiomer or it can be a mixture of both. Enantiomerically pure materials are referred to as *homochiral* or *enantiopure*. The 1:1 mixture of enantiomers has zero net rotation (because the rotations caused by the two enantiomers precisely cancel each other) and is called a *racemic mixture* or *racemate*. A racemic mixture has its own characteristic properties in the solid state. It differs in melting point and solubility from the pure enantiomers, owing to the fact that the racemic mixture can adopt a different crystalline structure from that of the pure enantiomers. For example, Figure 2.1 shows the differing intermolecular hydrogen-bonding and crystal-packing arrangements in (+/–) and (–) 2,5-diazabicyclo[2.2.2]octa-3,6-dione.¹

The composition of a mixture of enantiomers is given by the *enantiomeric excess*, abbreviated e.e., which is the percentage excess of the major enantiomer over the minor enantiomer:

$$\text{e.e.} = \% \text{ Major} - \% \text{ Minor} \quad (2.2)$$

Nomenclature of stereoisomers

Fisher Nomenclature



Emil Fisher
Nobel Prize in 1902

Used for biomolecules such as amino acids and sugars

Based on optical rotation of molecules

“+” → Right (clockwise) → Dextrorotatory → **D**

“-” → Left (anticlockwise) → Levorotatory → **L**

Fisher nomenclature

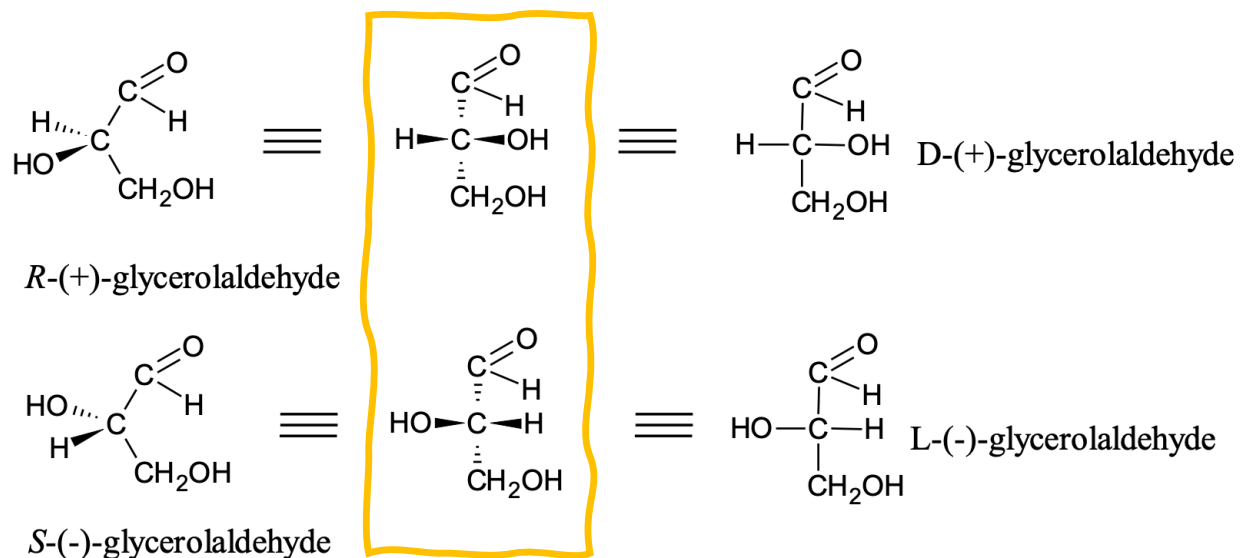
The group in the top position must be the carbon of highest priority

2. the projected formula and the Fischer's notation

rules of projecting:

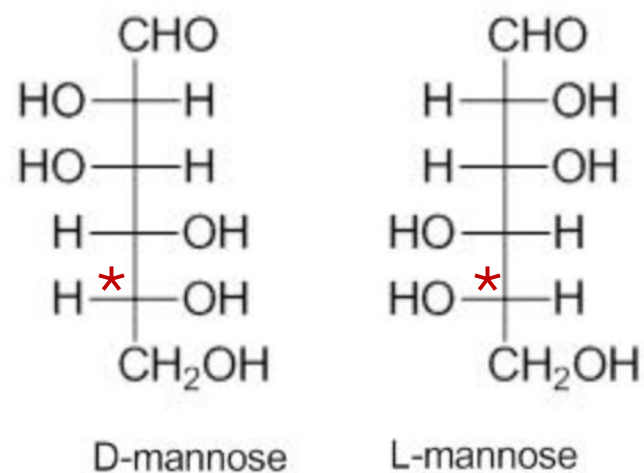
- a) the group on the highest oxidation level is found on the top
- b) the longest chain is down
- c) both previous groups are behind the plane of the vertical plane
- d) the other two groups are located horizontally, in front of the

plane of the vertical plane



Fisher nomenclature

In case of multiple chiral centers, the penultimate carbon (last to final) carbon/ the last chiral carbon to be considered



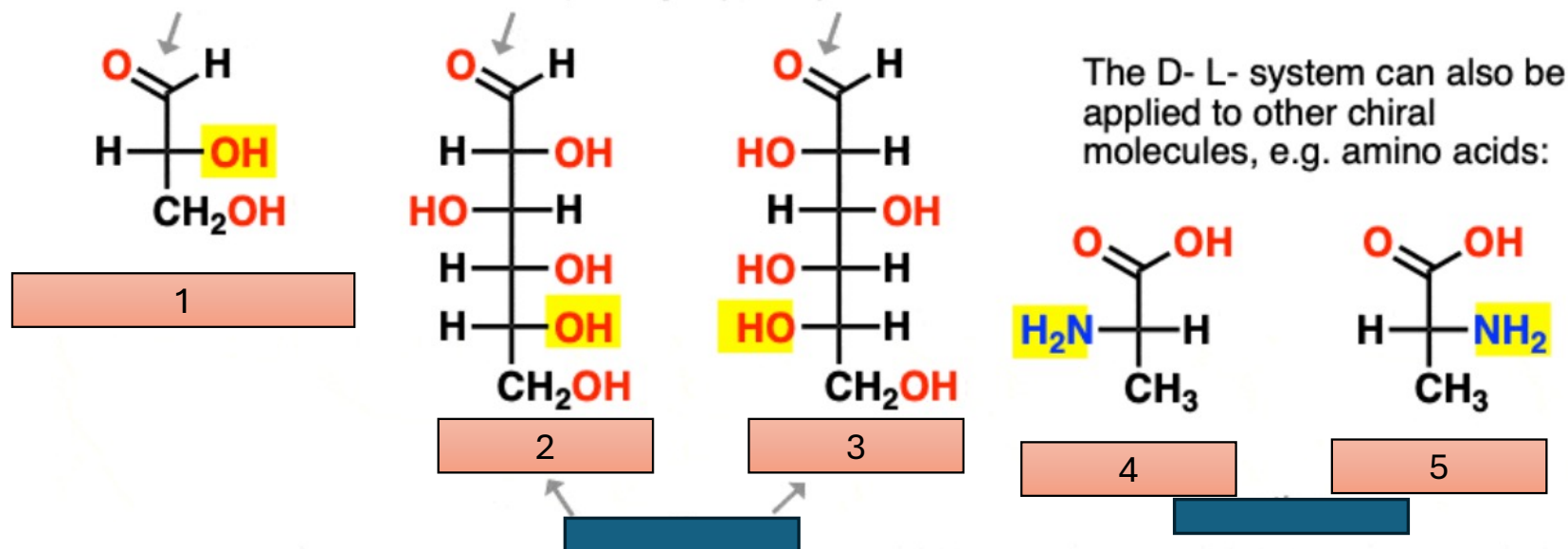
Fisher notation? Yes or NO

Summary: D- and L- Sugars

For a sugar drawn in the Fischer projection with the most oxidized carbon at the top:

- If the OH on the bottom chiral center points to the **right**, the sugar is **D**
- If the OH on the bottom chiral center points to the **left**, the sugar is **L**

most oxidized carbon (aldehyde) at top: these are "aldoses"



L- and D- is a means of describing the **absolute configuration** of a molecule that pre-dates *R* and *S* but is still used for some biological molecules (sugars, amino acids).

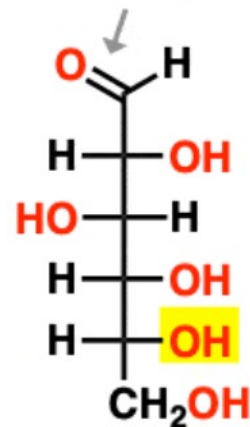
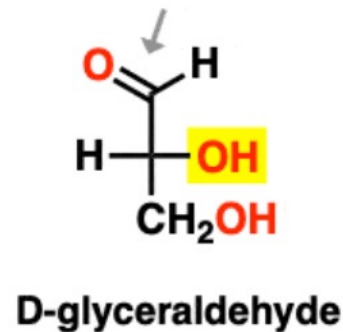
L- and D- have no relation to the optical rotation of a molecule.

Summary: D- and L- Sugars

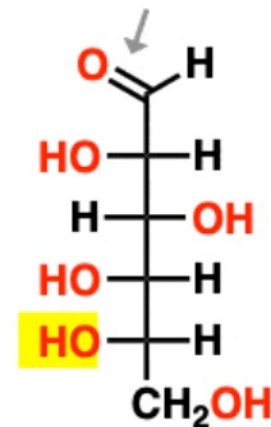
For a sugar drawn in the Fischer projection with the most oxidized carbon at the top:

- If the OH on the bottom chiral center points to the **right**, the sugar is **D**
- If the OH on the bottom chiral center points to the **left**, the sugar is **L**

most oxidized carbon (aldehyde) at top: these are "aldoses"



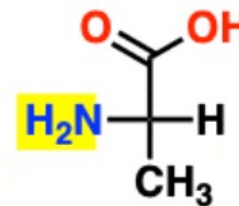
D-Glucose



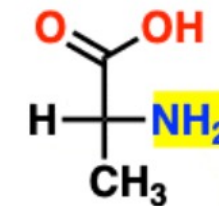
L-Glucose

enantiomers

The D- L- system can also be applied to other chiral molecules, e.g. amino acids:



L-alanine



D-alanine

enantiomers

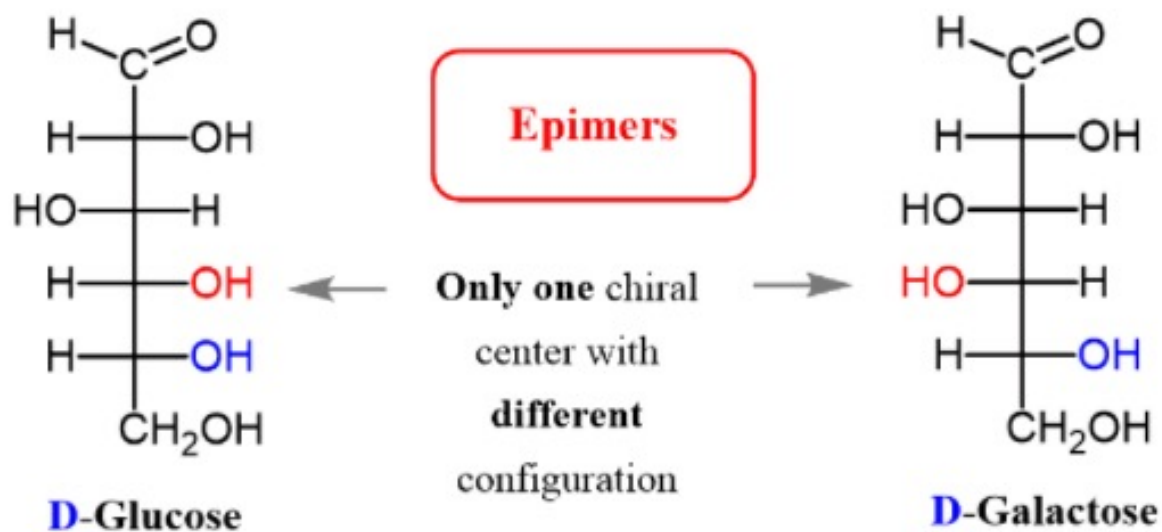
L- and D- is a means of describing the **absolute configuration** of a molecule that pre-dates *R* and *S* but is still used for some biological molecules (sugars, amino acids).

L- and D- have no relation to the optical rotation of a molecule.

Epimers

Epimers are molecules that differ in the configuration of only one stereogenic center out of at least two. They are a type of diastereomer, which means they have a similar structure but are not mirror images of each other.

D-glucose and D-galactose are epimeric at carbon-4

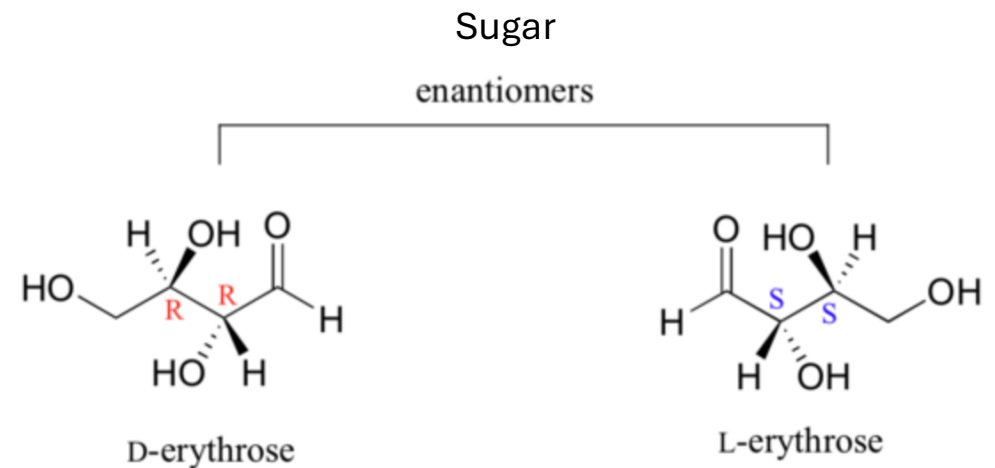
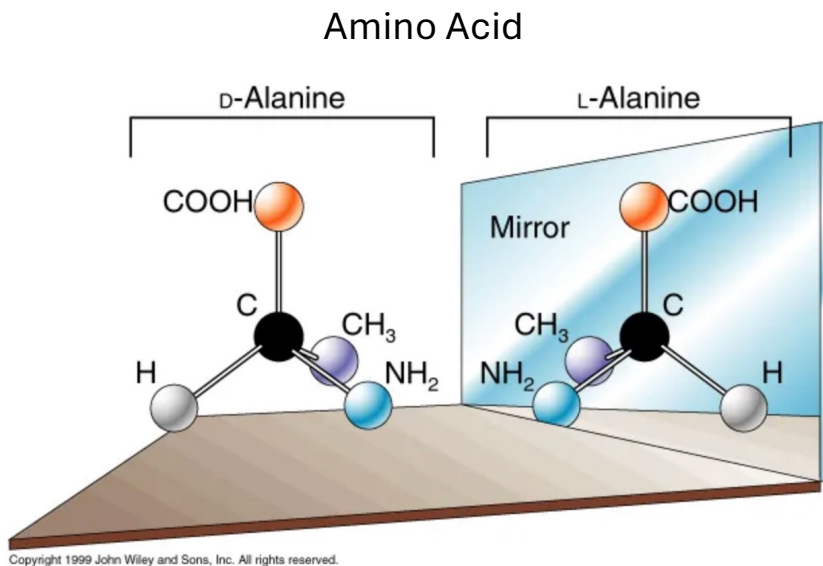


Glucose and mannose: These sugars are C2 epimers.

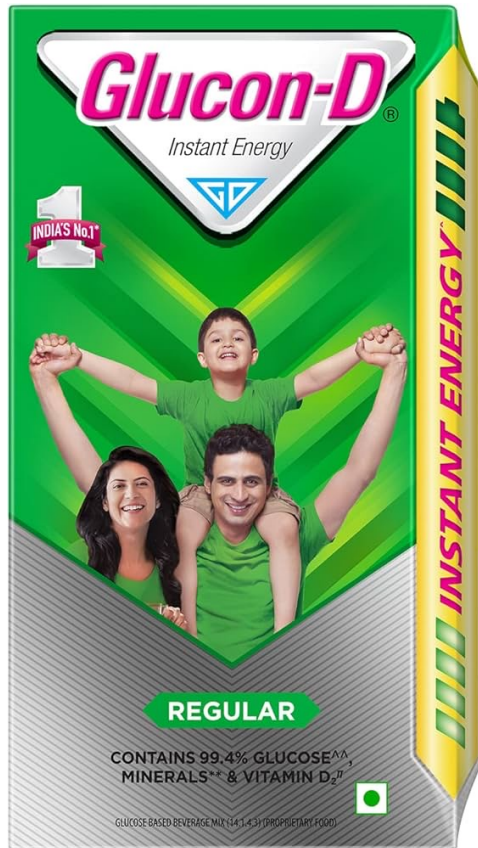
Ribose and xylose: These sugars are C3 epimers.

D-Arabinose and L-xylose: These sugars are C4 epimers.

Biological Homochirality: One of life's greatest mystery



Almost in every life form on Earth, sugars are always right-handed (D), and amino acids are always left-handed (L).



Glucon-L



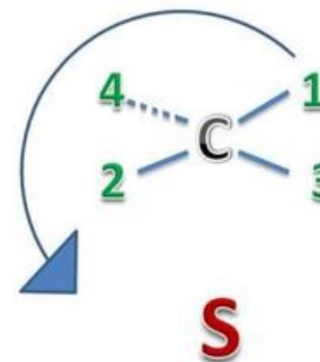
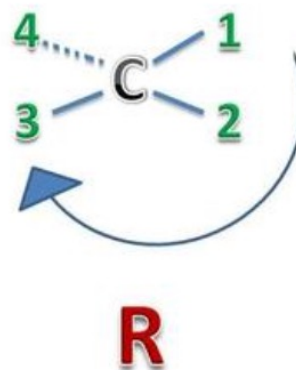
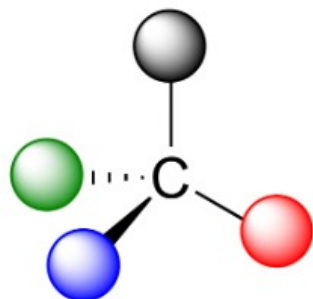
Nomenclature of stereoisomers

CIP Nomenclature

Cahn-Ingold-Prelog (CIP) nomenclature

Assign priorities to a chiral carbon center

Rotate the molecule, so that the lowest priority is on the dashed bond i.e. behind the plane



Sequence Rules to Assign Priorities to Substituents

Before applying the (*R*) and (*S*) nomenclature to a stereocenter, the substituents must be prioritized according to the following rules:

Rule 1

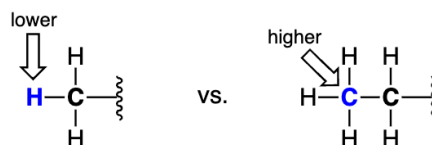
First, examine at the atoms directly attached to the stereocenter of the compound. A atom with a higher atomic number takes precedence over a atom with a lower atomic number. Hydrogen is the lowest possible priority atom, because it has the lowest atomic number.

1. The atom with higher atomic number has higher priority ($I > Br > Cl > S > P > F > O > N > C > H$).
2. When comparing isotopes, the atom with the higher mass number has higher priority [$^{18}O > ^{16}O$ or $^{15}N > ^{14}N$ or $^{13}C > ^{12}C$ or $T (^3H) > D (^2H) > H$].

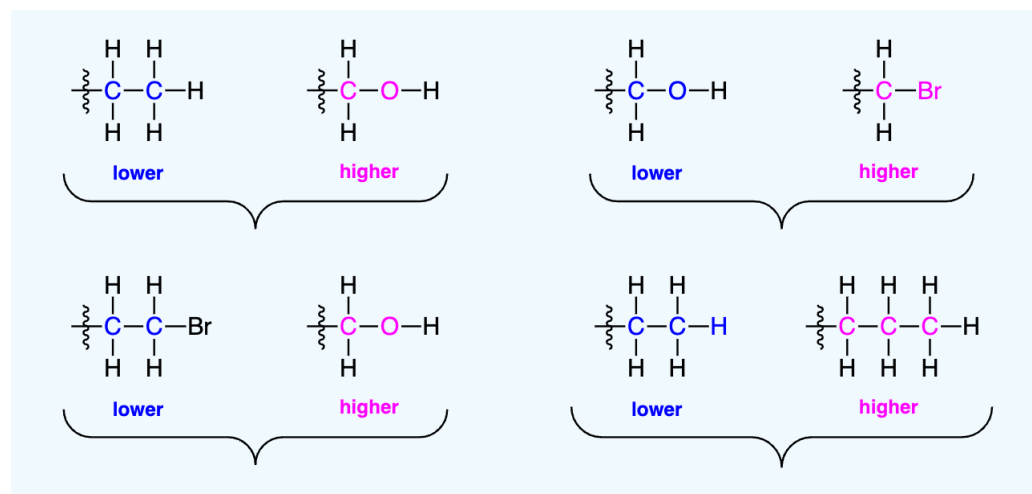
Rule 2

If there are two or more substituents which have the same element directly attached to chiral carbon, proceed along the substituent chains until a point of difference is found. Determine which of the chains has the first connection to an atom with the highest priority (the highest atomic number). That chain has the higher priority.

For example: an ethyl substituent takes priority over a methyl substituent. At the connectivity of the stereocenter, both have a carbon atom, which are equal in rank. Going down the chains, a methyl has only has hydrogen atoms attached to it, whereas the ethyl has two hydrogen atoms and a carbon atom. The carbon atom on the ethyl is the first point of difference and has a higher atomic number than hydrogen; therefore the ethyl takes priority over the methyl.

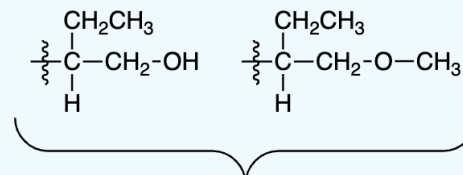
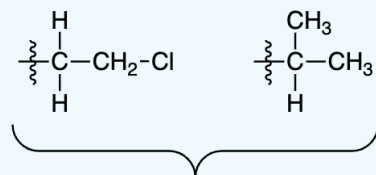
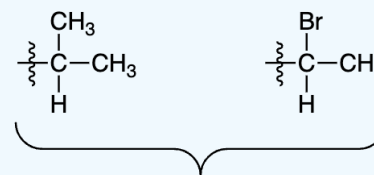
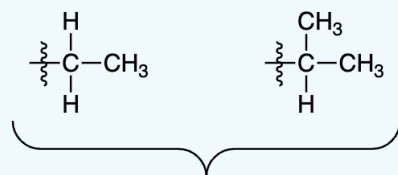


The "-H" (left) ranks lower than the "-C-" (right) based on the relative molecular weights at the first point of difference.



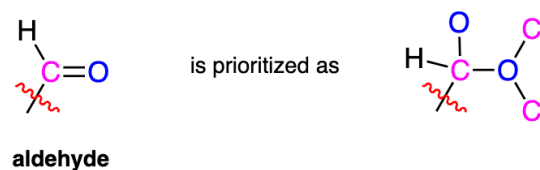
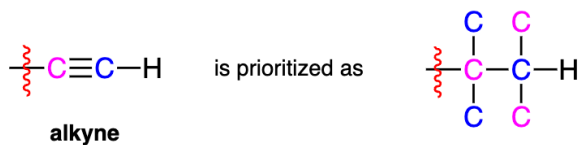
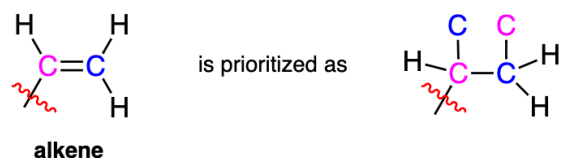
DIY

For the following pairs of substituents, determine which would have the higher and lower priority based on the **Cahn-Ingold-Prelog** rules. Explain your answer.



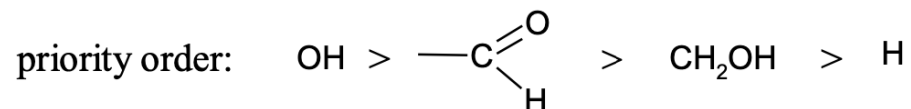
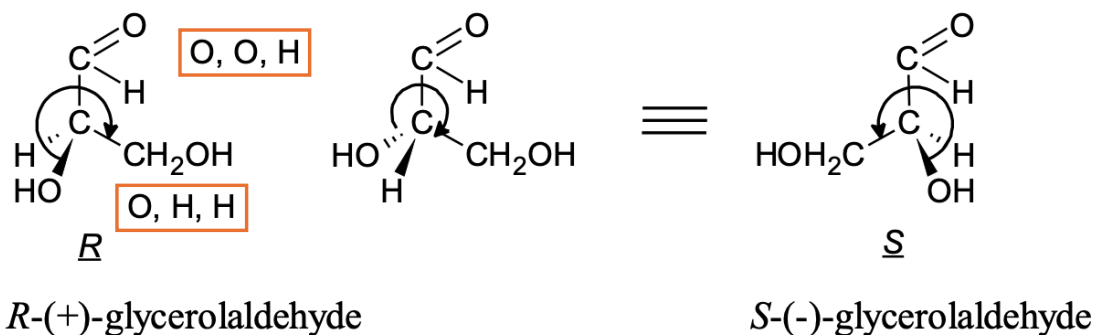
Rule 3

For assigning priority, multiple bonds are treated as if each bond of the multiple bond is bonded to a unique atom. For example, an alkene substituent ($\text{CH}_2=\text{CH}-$) has higher priority than an ethyl substituent (CH_3CH_2-). The alkene carbon priority is "two" bonds to carbon atoms and one bond to a hydrogen atom compared with the ethyl carbon that has only one bond to a carbon atom and two bonds to two hydrogen atoms. Similarly, alkyne substituent ($\text{HCC}-$) would have an even higher priority because the alkyne carbon is treated as if it is bonded to three carbons. This method remains the same with compounds containing a carbonyl ($\text{C}=\text{O}$) group. The carbon of an aldehyde substituent ($\text{O}=\text{CH}-$) is treated as if it is bonded to a hydrogen and two oxygen atoms.



Assigning *R* and *S*

1. perspective formula and the Cahn-Ingold-Prelog (CIP) notation system



rotate the group with the lowest priority to the farthest position from us, then the centre of chirality gets

R configuration, if the decreasing order of the other 3 groups follows clockwise direction

S configuration, if the decreasing order of the other 3 groups follows counterclockwise direction

R-(+)-glycerolaldehyde

S-(-)-glycerolaldehyde

advantage: absolutely clear

disadvantage: it does not represent the configurational relationship

Assigning CIP nomenclature

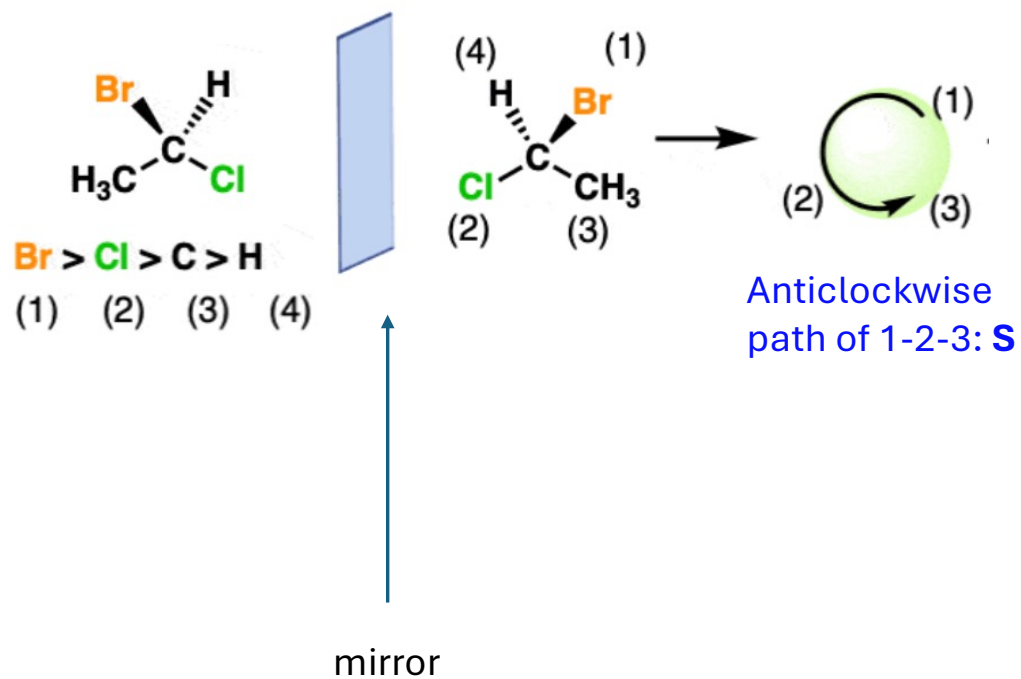
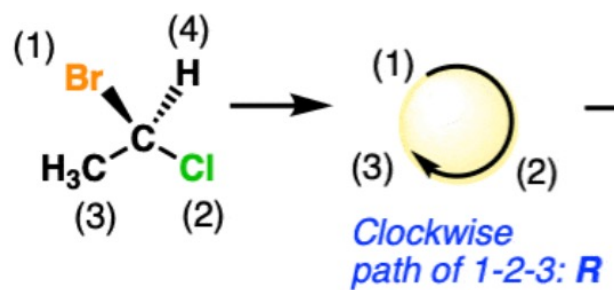
Step 1: Prioritize the four groups attached to the chiral center

Step 2: Orient the groups so that the lowest priority is on the dashed bond

Step 3: Determine if the sequence 1-2-3 is clockwise (R) or counterclockwise (S)

CIP nomenclature

Case 1: Lowest priority is on the dashed

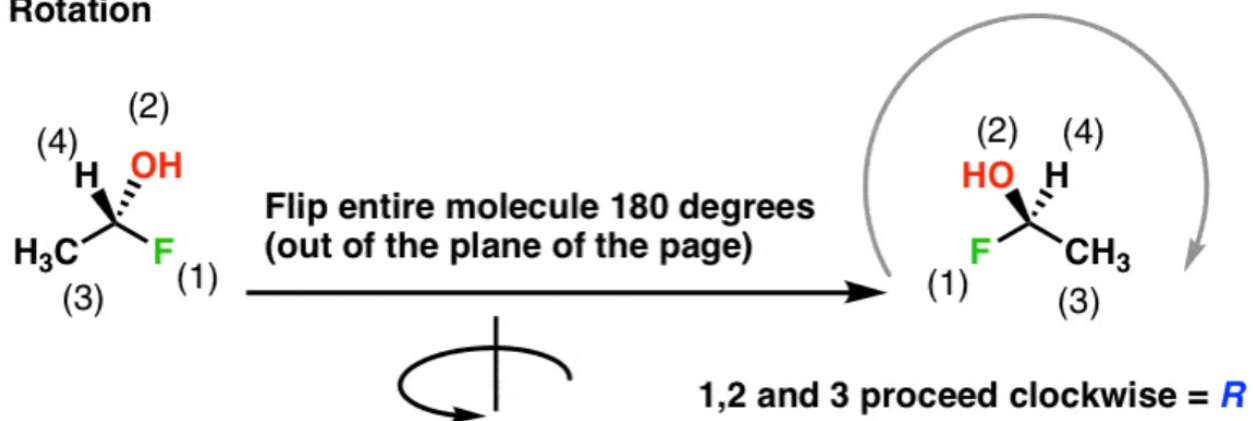


CIP nomenclature

Case 2: Lowest priority is on the wedge

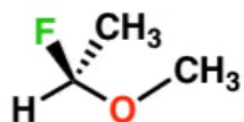


1. Rotation



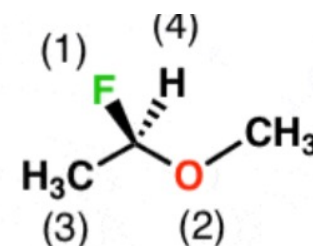
CIP nomenclature

Case 3: Lowest priority is on the plane



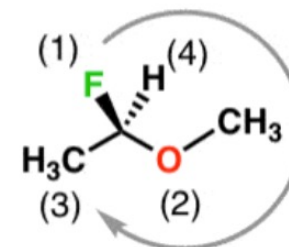
Step 1: Swap the priority #4 group (H) with whichever group is in the back (on a "dash")

[note: doing this will flip the configuration from R to S or vice versa. We correct for this in step 3]



single swap has occurred

Step 2: Determine (R)/(S) on the "swapped" chiral center.



*clockwise:
appears to be "R"*

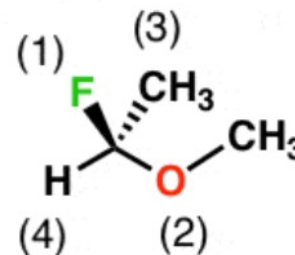
CIP nomenclature

Case 3: Lowest priority is on the plane

Step 3: Flip the result. Since we did a single swap, the "true" configuration of the chiral center is the opposite of what we obtained in Step 2.

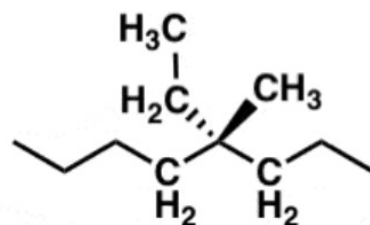
In this case, we flip our "R" result from step 2 to give the true configuration, (S).

→ Flip the result to the opposite: S



(S)-1-Fluoro-1-methoxyethane

Assign absolute configuration to this molecule



1. First assign priorities to the groups

2. Check where the lowest priority group lies?

Is it on a dashed bond?

Is it on a wedge bond?

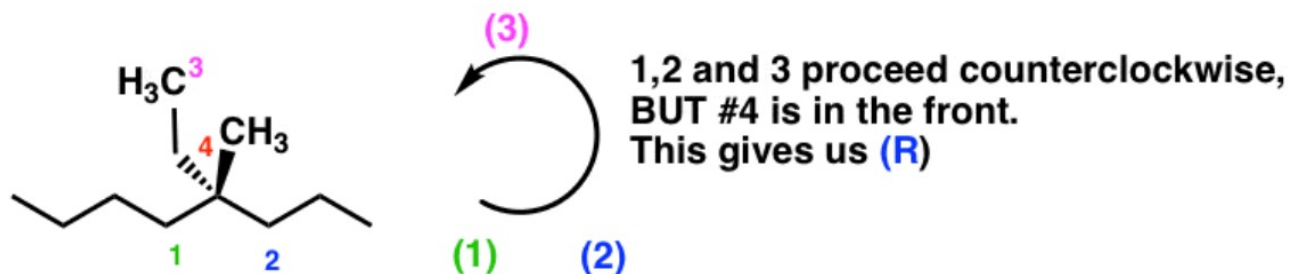
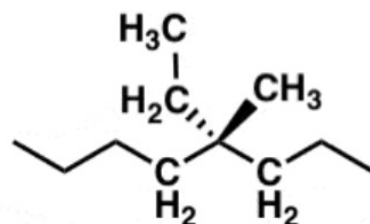
Is it on the plane?

If on the dash, check clockwise (R) or anticlockwise (S) and denote accordingly

If on the wedge, then flip the entire molecule, check clockwise (R) or anticlockwise (S) and denote accordingly

If on the plane, then swap and assign (R/S) and then swap the final configuration again (S/R)

Assign absolute configuration to this molecule



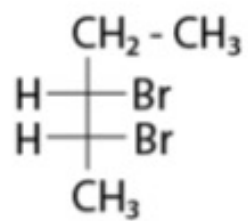
Flip the entire molecule because lowest priority is on the wedge bond

Diastereomers are stereoisomers (molecules with the same formula but different spatial arrangements) that are not mirror images of each other

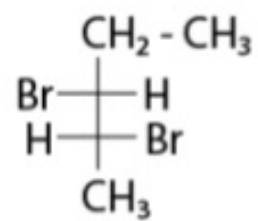
Unlike enantiomers,
Diastereomers have multiple chiral centers where configurations differ at some but not all.

They possess distinct physical properties (melting/boiling points, density)

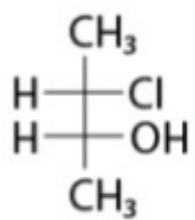
And sometimes different chemical reactivity, making them easier to separate than enantiomers, which have identical physical properties



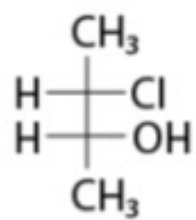
Erythro



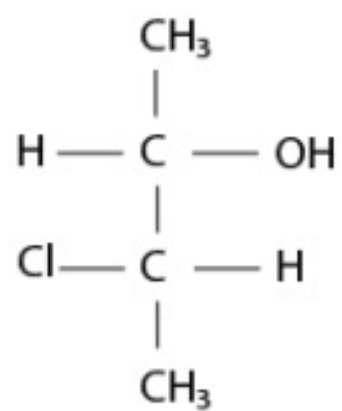
Threo



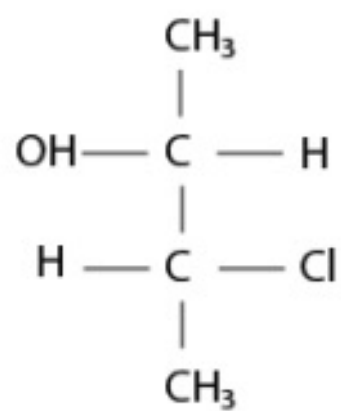
Erythro



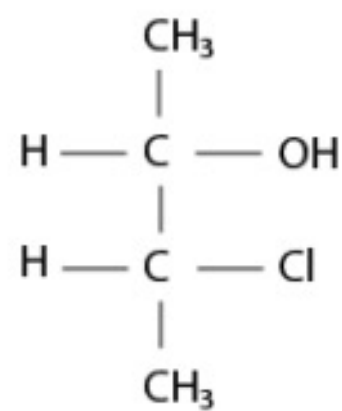
Threo



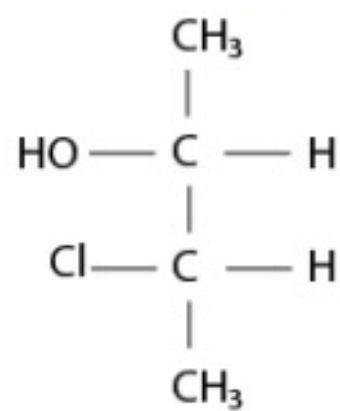
1



2



3



4

Isomers of 2,3,4-Trihydroxybutanal

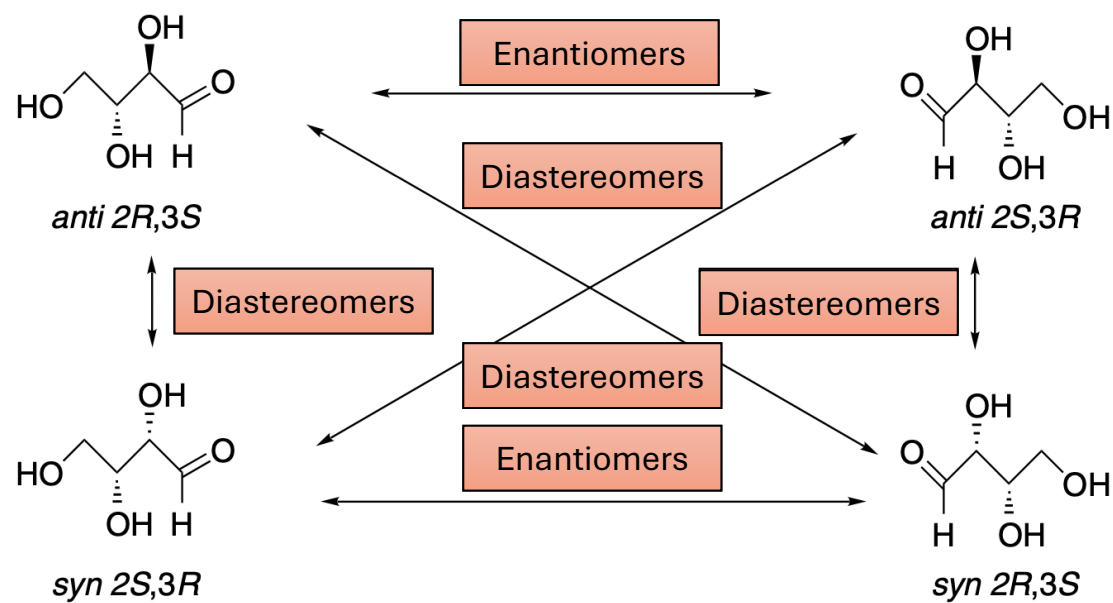
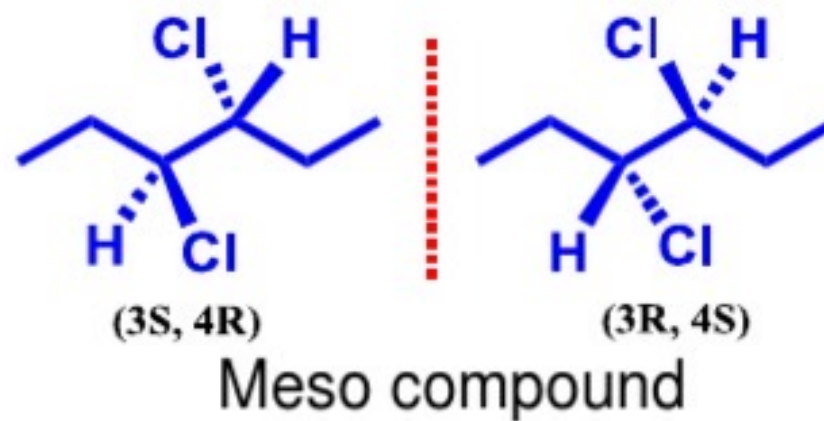


Fig. 2.4. Extended chain representation of all stereoisomers of 2,3,4-trihydroxybutanal.

Ref: Carey and Sundberg

Compounds
with chiral
centers can
also be
optically
inactive

Meso compound



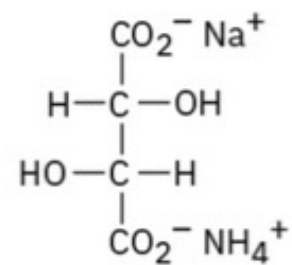
Plane of symmetry



Louis Pasteur

1848

Crystallization of tartaric acid



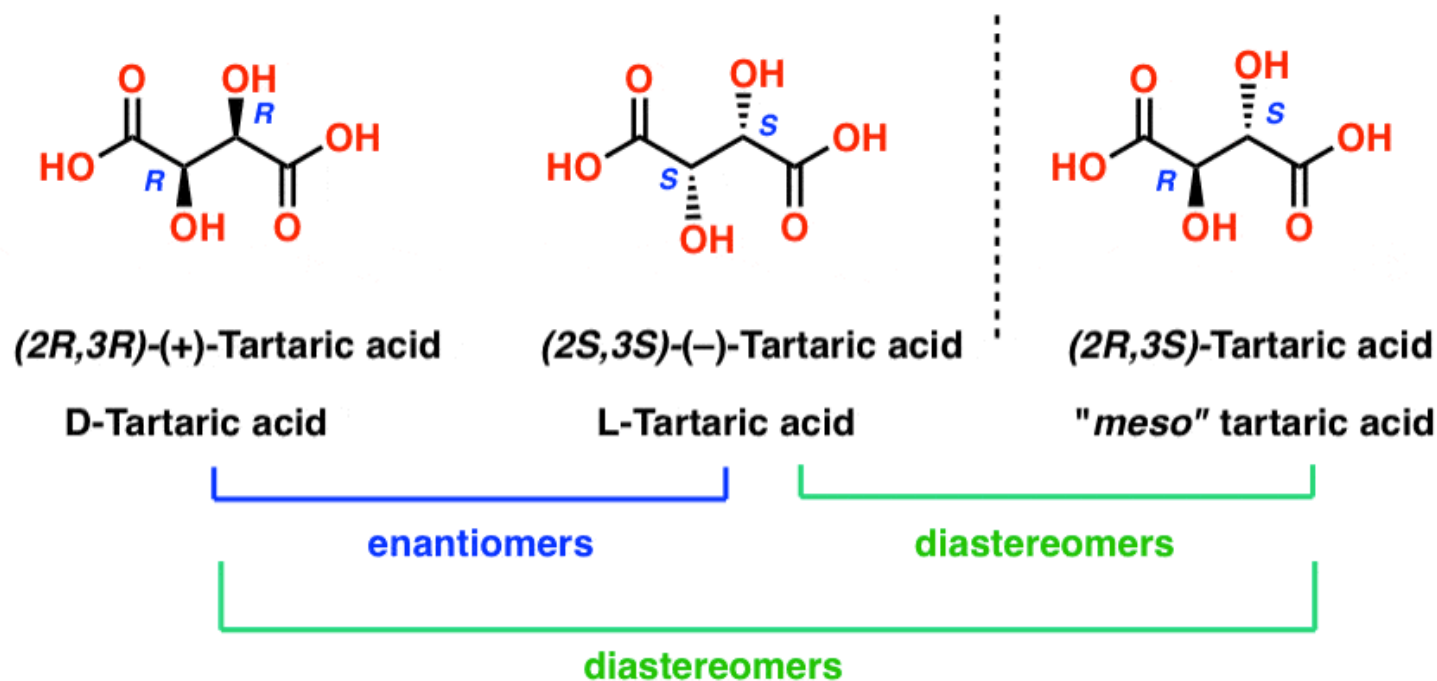
white

Sodium ammonium tartrate



How many isomers of Tartaric acid possible?

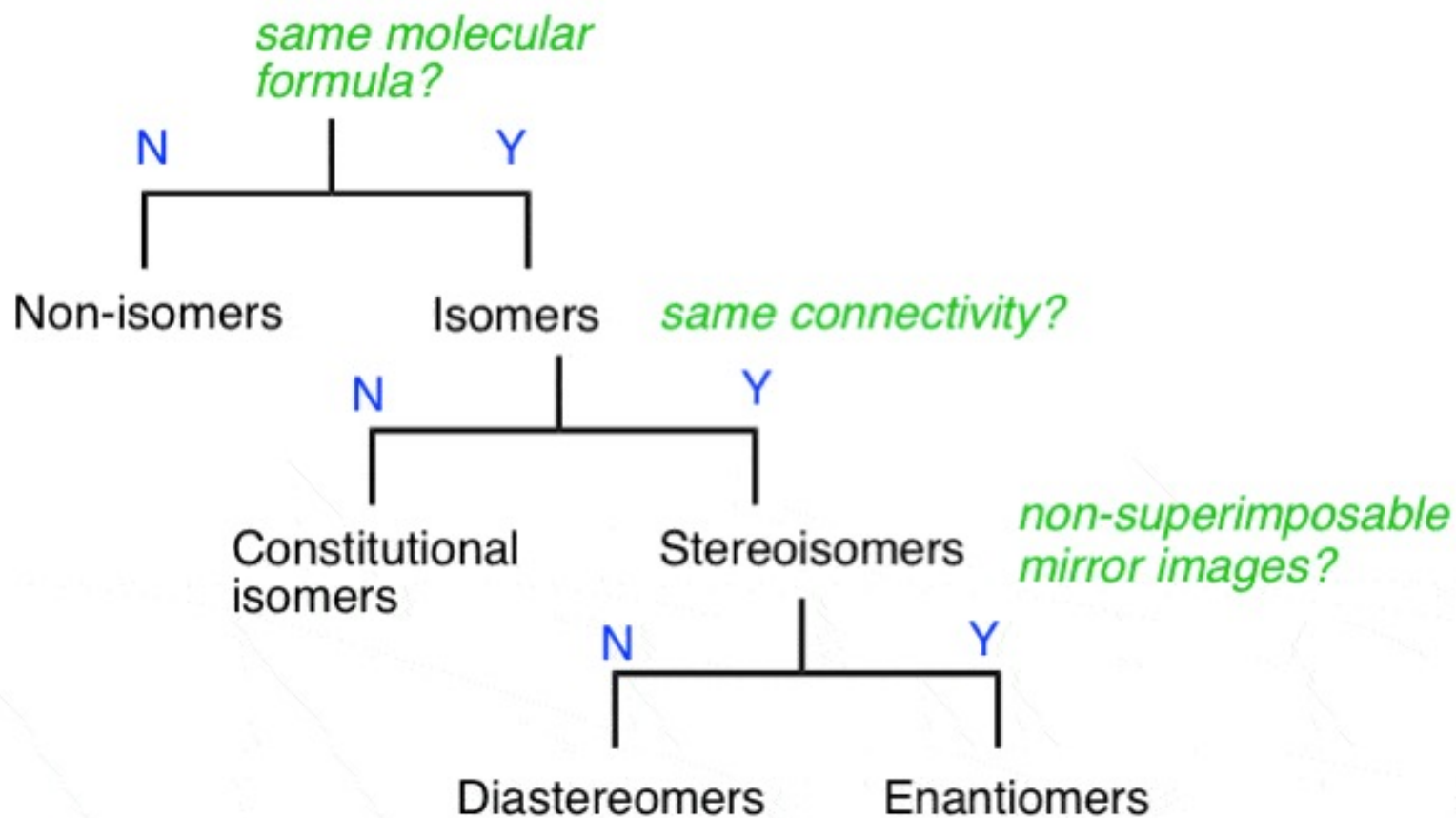
Three stereoisomers of tartaric acid (2,3-dihydroxysuccinic acid)



Can also form a racemic mixture (optically inactive)

Try to draw the fisher notation and assign D or L

Possible relationships between two molecules



Types of stereoisomerism

Conformational
Isomerism



Configurational
Isomerism



Optical
Isomerism

Enantiomers

Diastereomers

Geometrical
Isomerism

Cis/Trans
isomers

E/Z
isomers